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Do Hedge Funds Still Hedge?
A Pre-Registered Out-of-Sample Test of Asness, Krail and Liew (2001)

A replication and 20-year out-of-sample study of stale prices and hidden market exposure in hedge fund indices

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Abstract

Asness, Krail and Liew (2001) showed that hedge funds hedge much less than simple statistics suggest: because many funds hold illiquid or hard-to-price assets, their reported returns react to market moves with a delay, and adding lagged market returns to a regression more than doubled the market beta of the aggregate hedge fund index and turned its alpha negative. This study tests whether that conclusion still holds, using a pre-registered design in which the hypotheses, decision rules, a power analysis and 144 robustness specifications were fixed before any regression was run. With the free EDHEC hedge fund style indices and the market factor of the Kenneth French Data Library, the original result replicates for January 1997 to September 2000: the lagged beta of an equal-weighted composite is 0.20 ($p = 0.001$) and triples its measured market exposure. In the 248 months from October 2000 to May 2021, the primary out-of-sample test also supports the hypothesis, but the effect is much smaller: the lagged beta is 0.057 ($p = 0.009$), raising the measured beta from 0.14 to 0.20, and it is significant in 127 of the 144 specifications. The lag is concentrated in illiquid strategies, such as distressed securities and fixed-income arbitrage, and is also present in the Barclay Hedge Fund Index in 2022–2024. Unlike in the original study, however, the lag does not remove hedge fund alpha over the full period, and it is larger after rising markets than after falling ones. Hedge funds hedge more than they did before 2000, but still less than their reported returns suggest.

Keywords: hedge funds, market exposure, stale prices, lagged betas, return smoothing, replication, out-of-sample testing, pre-registration

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1. Introduction

The name "hedge fund" promises something specific: a portfolio whose gains do not depend on the direction of the stock market, because its exposure to that market has been hedged away. The history of the industry's most famous failures suggests that the promise is not always kept. Long-Term Capital Management lost most of its capital within a few weeks in 1998 when positions that looked unrelated on paper all moved against it at once, so that its hedges stopped working exactly when they were needed (Jorion, 2000). Amaranth Advisors lost roughly six billion dollars in September 2006 on a concentrated and largely unhedged bet on natural gas prices (Chincarini, 2007). In March 2021, Archegos Capital Management defaulted on highly leveraged and concentrated equity positions held through total return swaps, costing Credit Suisse alone about five and a half billion dollars (Paul, Weiss, Rifkind, Wharton & Garrison LLP, 2021). These collapses had different causes, including leverage, concentration and forced selling, but they share a common thread: the funds carried far more directional risk than their label implied, either because the risk was never hedged or because the hedges failed under stress.

Spectacular failures are rare, and they are not proof that hedge funds in general do not hedge. The more systematic question was asked by Asness, Krail and Liew (2001) in a paper titled simply "Do hedge funds hedge?". Using the CSFB/Tremont hedge fund indices from January 1994 to September 2000, they showed that the usual way of measuring market exposure, a regression of monthly hedge fund returns on contemporaneous stock market returns, understates the true exposure. Many hedge funds hold illiquid or hard-to-price securities, and their reported monthly returns adjust to market moves with a delay. When lagged market returns are added to the regression, the market beta of the aggregate hedge fund index more than doubles, from 0.37 to 0.84, and its apparent alpha of 2.6% per year turns into a loss of about 4.5% per year. Their answer to the title question was that hedge funds hedge far less than simple statistics suggest.

This study asks whether that conclusion still holds more than twenty years later. The hedge fund industry has changed since 2000: it has grown many times larger, become dominated by institutional investors, faced stricter valuation and reporting requirements after the 2008 financial crisis, and moved part of its assets into more liquid strategies. Each of these changes could plausibly have reduced the stale pricing that Asness et al. (2001) documented. On the other hand, the most illiquid strategies, such as distressed debt and credit arbitrage, remain central to the industry. Whether the lagged market exposure has survived is therefore an open empirical question with direct consequences for investors, who rely on reported betas, correlations and Sharpe ratios to decide how much of their portfolio to allocate to hedge funds.

The study is designed as a pre-registered replication and out-of-sample test. Before any regression was run, the hypotheses, data, samples, tests, decision rules, a power analysis and a grid of 144 robustness specifications were written down and fixed. The data are the freely

available EDHEC hedge fund style indices, which begin in January 1997, together with the market factor of the Kenneth French Data Library. The replication covers January 1997 to September 2000, the part of the original sample that free data allow. The primary test covers the 248 months from October 2000 to May 2021, all of which lie after the end of the original study. An extension uses the Barclay Hedge Fund Index for January 2022 to December 2024.

The results confirm the original finding but show that its size has shrunk. In the replication period, the lagged beta of the EDHEC composite is 0.20 and highly significant, tripling the measured market beta from 0.10 to 0.32. Out of sample, the lagged beta remains positive and significant (0.057, $p = 0.009$), and it is significant in 127 of the 144 robustness specifications, but it adds only about 40% to the measured beta, raising it from 0.14 to 0.20. The lag is concentrated in the least liquid strategies, as the stale-pricing explanation predicts. Unlike in the original study, however, the lagged exposure does not eliminate hedge fund alpha over the full period, and the lag is larger after rising markets than after falling ones, which is the opposite of the pattern that Asness et al. (2001) interpreted as evidence of deliberate smoothing. In 2022–2024, the Barclay index still shows a significant lagged beta, but its market exposure is dominated by an ordinary contemporaneous beta: its correlation with the stock market is 0.94.

The remainder of the paper is organised as follows. Chapter 2 reviews the literature on hedge fund risk and performance, on non-synchronous prices and the measurement of beta, and on return smoothing. Chapter 3 describes the pre-registered design, the data and the tests. Chapter 4 presents the results, Chapter 5 discusses what they mean and what they cannot show, and Chapter 6 concludes.

2. Literature Review

This chapter reviews the research on which the study builds. It first describes how hedge fund risk and performance are usually measured and the biases that affect hedge fund data. It then turns to the econometrics of stale prices and lagged betas, to the evidence that hedge fund returns are smoothed, and to the research gap addressed here.

2.1 Measuring hedge fund risk and performance

Early academic studies of hedge funds evaluated them in the same way as mutual funds, by regressing their returns on market indices. Ackermann et al. (1999) found that hedge funds outperformed mutual funds on a risk-adjusted basis, although not market indices, and that their returns were only weakly related to the stock market. Fung and Hsieh (1997) showed, however, that hedge fund returns have low and unstable correlations with standard asset classes because their strategies are dynamic: managers trade actively, use leverage and take short positions, so that their exposures change over time and are poorly captured by a single static beta. Agarwal and Naik (2004) went further and showed that many hedge fund strategies have payoffs resembling a short position in put options on the equity market, earning steady returns in normal times and suffering large losses in market declines. A low average beta can therefore hide substantial exposure to market crashes.

The data used to measure performance are themselves imperfect. Hedge funds report voluntarily to commercial databases, and funds that stop reporting, often because they performed badly, drop out of many datasets. Liang (2000) documented the resulting survivorship bias, and Fung and Hsieh (2000) distinguished it from the backfill bias created when funds join a database and add their earlier, typically favourable, history. Both biases raise the average returns of hedge fund indices. To limit the influence of such problems, Fung and Hsieh (2004) proposed benchmarking hedge funds against a small set of risk factors built from traded assets rather than against peer averages.

Whether hedge funds deliver alpha once these issues are taken into account remains disputed. Kosowski et al. (2007) used bootstrap and Bayesian methods and concluded that the performance of top hedge funds cannot be explained by luck alone. Fung et al. (2008) found that only a minority of funds of funds delivered persistent alpha between 1995 and 2004, and that capital inflows eroded it. Dichev and Yu (2011), measuring the returns actually earned by investors rather than buy-and-hold returns, found that dollar-weighted returns were three to seven percentage points lower than reported returns, because investors tended to add money after good performance. For the present study, the implication is that any alpha estimated from hedge fund indices is an upper bound on what investors earned.

2.2 Stale prices and lagged betas

The problem that Asness et al. (2001) applied to hedge funds was first studied for individual stocks. When a security trades infrequently, its last recorded price at the end of a

period may reflect information from earlier in the period, so its measured return responds to market news with a delay. Scholes and Williams (1977) showed that ordinary least squares then produces biased and inconsistent estimates of beta, downward for thinly traded securities, and proposed a consistent estimator combining leads and lags of the market return. Dimson (1979) proposed the simpler approach of regressing a security's return on the current and several lagged market returns and summing the coefficients. Both papers were motivated by small, illiquid stocks, whose simple betas were systematically too low.

Asness et al. (2001) argued that hedge fund returns suffer from the same problem, for two reasons. Some holdings, such as convertible bonds, distressed debt or over-the-counter derivatives, trade rarely and are marked at stale prices. Others are valued by the managers themselves, with the possibility that valuations are smoothed on purpose. Applying the regression of Dimson (1979) with three lags to the CSFB/Tremont indices, they found that lagged market returns enter significantly for the aggregate index and for most styles, most strongly for convertible arbitrage, event-driven and fixed-income arbitrage, the categories holding the least liquid securities. They also found that the lagged betas were much larger after down markets than after up markets, which they interpreted as circumstantial evidence of intentional smoothing, since a manager concerned with reported volatility has more reason to delay the recognition of losses than of gains. Finally, the Sharpe ratio of the aggregate index fell from 0.80 to −0.40 when the index was hedged with its summed beta rather than left unhedged.

2.3 Return smoothing and serial correlation

Stale or smoothed prices leave a second signature: positive serial correlation in reported returns. Getmansky et al. (2004) modelled reported returns as a weighted average of current and past true economic returns:

$$r_t^o = \theta_0 r_t + \theta_1 r_{t-1} + \ldots + \theta_k r_{t-k}, \quad \theta_0 + \theta_1 + \ldots + \theta_k = 1$$

where r^o_t is the reported return and r_t the true return. Such smoothing leaves the average return unchanged but reduces measured volatility and creates positive autocorrelation. Getmansky et al. (2004) found that hedge funds show far more serial correlation than mutual funds or stock indices, and argued that illiquidity, rather than market inefficiency, is the most likely explanation. Lo (2002) showed that this autocorrelation can overstate annualised Sharpe ratios by more than 65%, because scaling monthly volatility by the square root of twelve assumes independent returns.

Later work examined whether smoothing is deliberate. Bollen and Pool (2008) found that serial correlation in hedge fund returns is conditional: it is stronger when returns are negative, consistent with managers delaying the recognition of losses. Bollen and Pool (2009) documented a discontinuity in the pooled distribution of monthly hedge fund returns, with significantly fewer small losses than small gains, which they attributed to misreporting. Cassar and Gerakos (2011) showed that funds with weaker pricing controls, for example those whose

managers value their own positions, report smoother returns. Illiquidity also affects investors directly. Aragon (2007) showed that funds with lock-up periods and redemption restrictions earn higher returns, interpreted as compensation for illiquidity, and hold less liquid assets. Merger arbitrage provides an example of a genuine delayed relation to the market: Mitchell and Pulvino (2001) found that merger arbitrage returns resemble short put positions on the market, and that deal failures become more likely after market declines, which could produce lagged betas even without stale prices.

2.4 Research gap

Asness et al. (2001) ended their sample in September 2000, after a period dominated by the late-1990s bull market and the 1998 crisis. Since then, the hedge fund industry has grown and institutionalised, valuation practices have tightened, and two major market crises, in 2008 and 2020, have tested the funds' exposures. The smoothing literature has studied serial correlation and misreporting, but the specific claim of Asness et al. (2001), that lagged market returns reveal hidden market exposure and remove apparent alpha, has not been tested on a clean out-of-sample period under a protocol fixed in advance. Without such a protocol, the choice of lags, indices, periods and standard errors leaves room to find either confirmation or refutation (Nosek et al., 2018). This study fills that gap with a pre-registered replication, a 20-year out-of-sample test, a three-year extension on a different data source, and a complete report of all pre-specified robustness checks.

3. Methodology

This chapter describes the pre-registered design: the hypotheses, the data, the regression model, the decision rules and the power analysis, and the robustness and verification checks.

3.1 Pre-registered design

The study was pre-registered: the hypotheses, samples, test statistics, decision rules, power analysis and robustness grid were written down before any regression, correlation or summary statistic was computed. Only three things had been seen at that point: the first and last rows of the EDHEC data file, checked to confirm that it loaded; the public table of the Barclay Hedge Fund Index, seen while checking which data were freely available; and the original study, whose estimates were used to set the power analysis. No deviation from the plan was made. Table 1 lists the six hypotheses. H2, the out-of-sample test of the lagged beta, is the single primary test; the others are secondary and are reported whatever they show.

Table 1. Pre-registered hypotheses.

	Hypothesis	Index and sample	Test
H1	Replication: the lagged beta is positive	EDHEC composite, January 1997 – September 2000	One-sided, lagged beta > 0
H2	Primary: the lagged beta is still positive out of sample	EDHEC composite, October 2000 – May 2021	One-sided, lagged beta > 0
H3	Once the lagged exposure is counted, hedge funds show no alpha	EDHEC composite, October 2000 – May 2021	95% interval of the lagged-model alpha
H4	Illiquid strategies show more lagged beta than liquid ones	Illiquid minus liquid portfolio, October 2000 – May 2021	One-sided, lagged beta of the difference > 0
H5	The lagged beta is still present in 2022–2024	Barclay Hedge Fund Index, January 2022 – December 2024	One-sided, lagged beta > 0
H6	The lag is larger after down markets than after up markets	EDHEC composite, October 2000 – May 2021	One-sided, down minus up > 0

Note: H2 is the single primary test; the others are secondary. Illiquid: Convertible Arbitrage, Distressed Securities, Event Driven, Fixed Income Arbitrage, Relative Value. Liquid: CTA Global, Equity Market Neutral, Global Macro, Long/Short Equity. Source: Author's own calculations.

3.2 Data

Hedge fund returns come from two sources. The main source is the EDHEC hedge fund style indices, monthly net-of-fee returns for 12 strategies and for funds of funds from January 1997 to May 2021, as distributed with the PerformanceAnalytics software package (Peterson & Carl, n.d.). The EDHEC indices are built as "indices of indices": each style index combines the competing commercial indices of that style, which reduces the influence of any single data provider's selection and biases (Goltz et al., 2007). The main dependent variable is the EDHEC composite, the equal-weighted average of the 12 style indices. Funds of funds are excluded from the composite to avoid counting the same funds twice, and are analysed separately. The distributor zero-filled seven unreported months of the Short Selling index between April 2020

and May 2021; these are treated as missing, and the composite averages the indices available in those months.

The extension uses the Barclay Hedge Fund Index, the average net return of all hedge funds, excluding funds of funds, in the BarclayHedge database (BarclayHedge, n.d.). Its full history is not freely available; the months from January 2022 to December 2024 were transcribed from the public index page, and the transcription was checked against the annual returns shown on the same page. The months from June to December 2021 are covered by neither free source.

Market returns and the risk-free rate come from the Kenneth French Data Library (French, n.d.). The market return is the value-weighted return of all CRSP firms in excess of the one-month T-bill. Asness et al. (2001) used the S&P 500; the broader CRSP index is a pre-declared deviation made because it is freely available, and the two are very highly correlated. The size, value and momentum factors (Fama & French, 1993; Carhart, 1997) from the same library are used as controls in the robustness grid. All hedge fund returns are converted to excess returns by subtracting the T-bill rate, and no observation after December 2024 is used. Table 2 reports summary statistics.

Table 2. Summary statistics of monthly excess returns.

Series	Months	Mean (%/yr)	Volatility (%/yr)	Sharpe ratio	Autocorrelation (lag 1)	Correlation with market
January 1997 – September 2000						
EDHEC composite	45	6.19	3.53	1.76	0.33	0.48
EDHEC Funds of Funds	45	9.98	7.95	1.26	0.22	0.73
Market (CRSP)	45	15.50	17.60	0.88	−0.12	1.00
October 2000 – May 2021						
EDHEC composite	248	3.92	3.69	1.06	0.29	0.61
EDHEC Funds of Funds	248	2.30	4.90	0.47	0.25	0.74
Market (CRSP)	248	7.47	15.64	0.48	0.10	1.00
January 2022 – December 2024						
Barclay Hedge Fund Index	36	−0.38	6.55	−0.06	−0.17	0.94
Market (CRSP)	36	5.73	17.74	0.32	−0.23	1.00

Note: Returns in excess of the one-month T-bill. Means and volatilities annualised by multiplying by 12 and √12. The EDHEC composite is the equal-weighted average of the 12 EDHEC style indices. Source: Author's own calculations.

Two features of Table 2 stand out before any regression. First, the hedge fund indices are highly autocorrelated: the first-order autocorrelation of the EDHEC composite is 0.33 in 1997–2000 and 0.29 in 2000–2021, against −0.12 and 0.10 for the market. Among the individual styles, it reaches about 0.50 for convertible arbitrage and fixed-income arbitrage and 0.45 for distressed securities, exactly the pattern that stale or smoothed prices produce. Second, the Barclay index behaved very differently in 2022–2024 from the EDHEC composite in earlier years: its volatility was 6.6% and its correlation with the market 0.94.

3.3 Model

Following Asness et al. (2001) and Dimson (1979), each hedge fund excess return is regressed on the current and three lagged market excess returns:

$$r_t = \alpha + \beta_0 m_t + \beta_1 m_{t-1} + \beta_2 m_{t-2} + \beta_3 m_{t-3} + \varepsilon_t$$

where r_t is the hedge fund excess return and m_t the market excess return in month t. Two quantities summarise the market exposure:

$$\text{Lagged beta} = \beta_1 + \beta_2 + \beta_3 \qquad \text{Summed beta} = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

The lagged beta measures the exposure that appears in reported returns only in later months; under stale or smoothed pricing it is positive. The summed beta is the full market exposure. Both are compared with the simple beta from the conventional regression on the current market return alone, estimated over the same months:

$$r_t = \alpha + \beta m_t + \varepsilon_t$$

Alphas are annualised by multiplying the monthly intercept by 12. Inference uses Newey and West (1987) standard errors with three lags, which allow for the autocorrelation documented above; the ordinary least squares tests used in the original study are also reported. For H6, each lagged market return is split into an up-market term and a down-market term according to its sign, giving six lag coefficients, as in Asness et al. (2001, Exhibit 5).

3.4 Decision rules and statistical power

All tests are one-sided at the 5% level against the pre-registered alternative, except H3, which is judged by a two-sided 95% confidence interval. A hypothesis is labelled Supported if its p-value is below 0.05. A non-significant result is labelled Not supported if the upper end of the 90% confidence interval is below 0.10, the smallest lagged beta considered economically meaningful, and Inconclusive otherwise. This three-way rule distinguishes an effect that is absent from an effect that the data are too short to detect. H3 is labelled Supported if the 95% interval of the lagged-model alpha includes or lies below zero, and Not supported if it lies entirely above zero.

Table 3 reports the power analysis carried out before the data were analysed. With 248 months, the primary test can detect a lagged beta of 0.09 with 80% power under the middle assumption for residual volatility, about a fifth of the 0.44 found for the aggregate index by Asness et al. (2001). The replication and the extension, with 45 and 36 months, have good power only for effects of about 0.2 and above, so non-significant results in those samples would more likely be Inconclusive than Not supported.

Table 3. Power analysis: minimum detectable lagged beta at 80% power (one-sided 5% test).

Test	Months	Residual volatility 1.0%	1.5%	2.3%	Power if true value is 0.10 (1.5%)
H1 replication	45	0.14	0.21	0.33	31%
H2 primary	248	0.06	0.09	0.14	86%
H5 extension	36	0.16	0.24	0.37	27%

Note: Computed before any regression was run. The standard error of the lagged beta is approximated by $\sigma\varepsilon\sqrt{3} / (\sigma m\sqrt{T})$, with a market volatility σm of 4.5% a month and three assumed monthly residual volatilities $\sigma\varepsilon$; 2.3% corresponds to the aggregate index of Asness et al. (2001). Source: Author's own calculations.

3.5 Robustness and verification

The primary test was repeated over a grid of 144 pre-specified specifications that vary five choices: the number of lags (one, two, three or six), the standard errors (ordinary least squares or Newey–West with three or six lags), the index (the EDHEC composite or the EDHEC Funds of Funds index), the sample (full, without September–December 2008, or without February–April 2020) and the controls (none, or the size, value and momentum factors). Every specification is reported. As exploratory analyses, the regressions were also run for each style index, with Holm (1979) adjusted p-values across the 12 styles, for the sub-periods before and after 2009, and over rolling 60-month windows.

Three checks verify the primary result. The primary test was recomputed by a second, independent implementation that reads the raw files by a different route and computes the regression and the Newey–West covariance matrix directly, without the statistical library (Seabold & Perktold, 2010) used in the main analysis; the two agree to within 10^{-10}. A look-ahead check confirms that each regression uses only current and past market returns. Finally, the data checks confirm that the zero-filled Short Selling months fall on the dates documented by the distributor, which also verifies the dating of the EDHEC file, and that no observation after December 2024 enters any calculation.

4. Results

This chapter presents the results in the order of the pre-registered hypotheses: the replication, the primary out-of-sample test, alpha, the role of liquidity, the 2022–2024 extension and the asymmetry test, followed by the robustness grid. Table 4 summarises all six decisions.

Table 4. Results of the pre-registered tests.

	Test	Estimate	Std. error	p-value	Interval	Decision
H1	Replication: lagged beta, composite, 1997–2000	0.203	0.066	0.001	[0.094, 0.311]	Supported
H2	Primary: lagged beta, composite, 2000–2021	0.057	0.024	0.009	[0.017, 0.097]	Supported
H3	Lagged-model alpha, composite, 2000–2021 (%/yr)	2.45%	0.90%	0.007	[0.68, 4.21]%	Not supported
H4	Lagged beta, illiquid minus liquid, 2000–2021	0.076	0.030	0.006	[0.027, 0.126]	Supported
H5	Lagged beta, Barclay index, 2022–2024	0.055	0.031	0.039	[0.004, 0.106]	Supported
H6	Down minus up lagged beta, composite, 2000–2021	−0.183	0.051	1.000	[−0.267, −0.099]	Not supported

Note: One-sided p-values against the pre-registered alternative (positive lagged beta; H6: down minus up > 0); for H3 the p-value is two-sided and the interval is the 95% interval, otherwise the 90% interval. Newey–West standard errors with 3 lags. Decision rules as pre-registered: Supported if $p < 0.05$; Not supported if $p \geq 0.05$ and the upper end of the 90% interval is below 0.10; otherwise Inconclusive. H3 is Supported if the 95% interval of the alpha includes or lies below zero. Source: Author's own calculations.

4.1 Replication, 1997–2000

The original result replicates. Over January 1997 to September 2000, the lagged beta of the EDHEC composite is 0.20 with a Newey–West standard error of 0.066 ($p = 0.001$), and H1 is supported. Adding the lagged market returns triples the measured market exposure, from a simple beta of 0.10 to a summed beta of 0.32. Table 5 compares the estimates with those of Asness et al. (2001) for the styles common to both studies.

Table 5. Replication: simple, lagged and summed betas in 1994–2000 (original) and January 1997 – September 2000 (this study).

	Asness et al. (2001), 1994–2000			This study, 1997–2000			
Index	Simple beta	Lagged beta	Summed beta	Simple beta	Lagged beta	Summed beta	p (lagged > 0)
Aggregate / composite	0.37	0.44	0.84	0.10	0.20	0.32	0.001
Convertible Arbitrage	0.04	0.35	0.43	0.06	0.24	0.32	0.005
Event Driven	0.28	0.30	0.61	0.29	0.25	0.57	0.006
Equity Market Neutral	0.12	0.08	0.20	0.09	0.08	0.18	0.004
Fixed Income Arbitrage	0.02	0.31	0.36	−0.01	0.34	0.36	0.011
Long/Short Equity	0.55	0.42	0.99	0.37	0.19	0.58	<0.001
Emerging Markets	0.74	0.46	1.25	0.74	0.41	1.19	0.055

Index	Asness et al. (2001), 1994–2000			This study, 1997–2000			
	Simple beta	Lagged beta	Summed beta	Simple beta	Lagged beta	Summed beta	p (lagged > 0)
Global Macro	0.37	0.57	0.98	0.29	0.27	0.58	0.002
CTA Global	0.01	−0.17	−0.19	0.01	−0.02	−0.02	0.558
Short Selling	−0.99	−0.25	−1.27	−1.21	0.08	−1.13	0.372

Note: Lagged beta = β1 + β2 + β3 from a regression on the current and three lagged market excess returns; summed beta = β0 + β1 + β2 + β3. The original study uses the CSFB/Tremont indices (aggregate index in the first row, whose closest counterpart here is the EDHEC composite; CTA Global is compared with Managed Futures and Short Selling with Dedicated Short Bias) and the S&P 500; this study uses the EDHEC indices and the CRSP market. One-sided p-values from Newey–West standard errors (3 lags). Source: Author's own calculations.

The pattern across styles is close to the original. The lagged betas are large and significant for convertible arbitrage (0.24), distressed securities (0.31), event driven (0.25), fixed-income arbitrage (0.34), global macro (0.27) and long/short equity (0.19), and close to zero for CTA and short-selling funds, as in the original study. Seven of the 12 style indices keep a significant lagged beta after the Holm correction (Appendix B). The levels differ, however. The EDHEC composite has a much lower beta than the CSFB/Tremont aggregate, 0.10 against 0.37, because it weights market-neutral and arbitrage styles equally with directional ones, whereas the aggregate index is weighted by assets, so that large directional styles such as long/short equity and global macro carry more weight. Relative to its simple beta, the composite's hidden exposure is at least as large as in the original: the summed beta is more than three times the simple beta, against slightly more than twice for the aggregate index. As in the original study, accounting for the lags removes most of the apparent alpha: the composite's alpha falls from 4.71% to 1.09% per year, and its Sharpe ratio after hedging with the summed beta falls from 1.76 to 0.26 (Appendix C).

4.2 Out of sample, 2000–2021

The primary test also supports the hypothesis, but with a much smaller effect. Over the 248 months from October 2000 to May 2021, the lagged beta of the EDHEC composite is 0.057 with a standard error of 0.024 ($p = 0.009$); the ordinary least squares test used in the original study gives $p = 0.002$. H2 is supported. The effect is less than a third of the replication estimate, and its 90% confidence interval, from 0.017 to 0.097, lies entirely below the smallest effect of interest of 0.10 set in the pre-registration. The lagged exposure is therefore real but modest: it raises the composite's measured beta from 0.14 to 0.20, an increase of about 40%. Table 6 shows the individual coefficients: nearly all of the lag comes from the first month, with $\beta_1 = 0.05$ and $\beta_2 = 0.02$, while the third lag is zero.

Table 6. Hedge fund market exposure out of sample: simple and lagged regressions.

Index and period	Simple beta	β0	β1	β2	β3	Lagged beta	Summed beta	Simple alpha (%/yr)	Lagged-model alpha (%/yr)
EDHEC composite, 2000–2021	0.14	0.14	0.05	0.02	−0.01	0.06	0.20	2.84	2.45
2000–2008	0.11	0.10	0.05	0.03	0.00	0.08	0.18	3.68	4.06
2009–2021	0.17	0.18	0.06	0.02	−0.01	0.07	0.25	1.69	0.49
EDHEC Funds of Funds, 2000–2021	0.23	0.23	0.06	0.03	−0.01	0.08	0.31	0.57	0.03
Barclay index, 2022–2024	0.35	0.36	0.06	0.00	−0.01	0.05	0.42	−2.37	−2.90

Note: Monthly excess returns regressed on the current (β0) and three lagged (β1–β3) CRSP market excess returns. Simple beta and alpha from the regression on the current market return only, over the same months. Alphas annualised (×12). 2000–2021 = October 2000 – May 2021; sub-periods split at December 2008. Source: Author's own calculations.

The lag is present in both halves of the period. Its point estimate is 0.08 between October 2000 and December 2008 and 0.07 between 2009 and May 2021, both significant at the 5% level. The EDHEC Funds of Funds index shows a similar lagged beta of 0.08 ($p = 0.001$), and a higher summed beta of 0.31, since funds of funds allocate more to directional strategies. Figure 1 compares the simple and summed betas of each style. The gap is largest for distressed securities, whose beta rises from 0.26 to 0.43, and for event-driven, fixed-income arbitrage and emerging markets funds, and it is essentially zero for global macro, merger arbitrage and CTA funds. After the Holm correction, the lagged betas of distressed securities, event driven and fixed-income arbitrage remain significant (Appendix A).

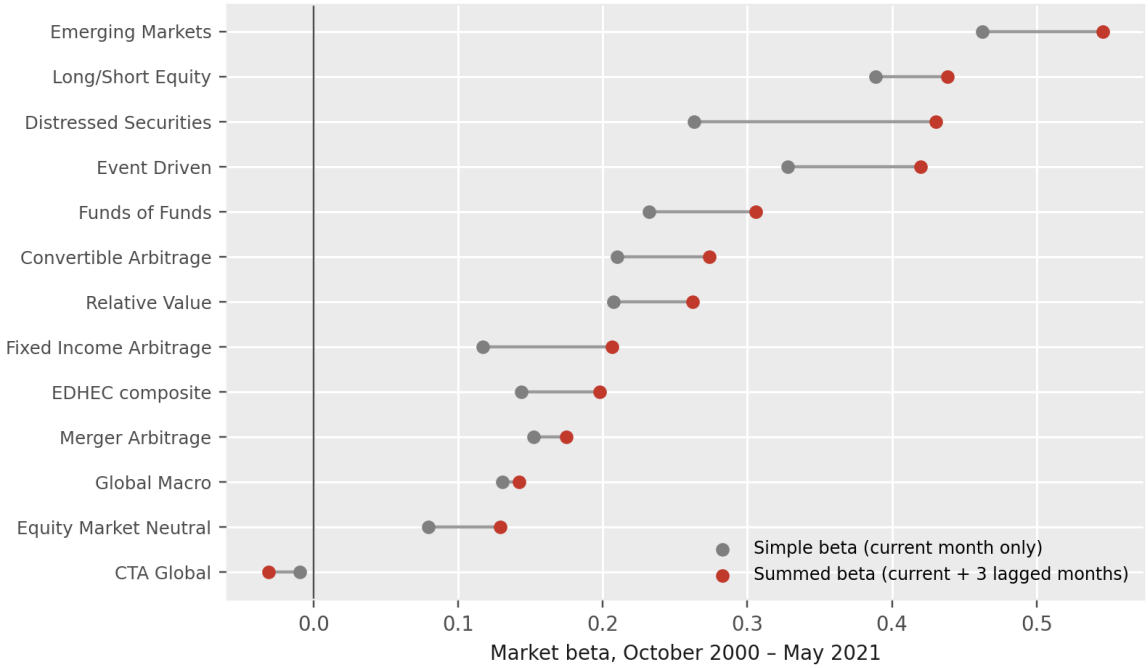


Figure 1. Simple and summed market betas of the EDHEC style indices, October 2000 – May 2021. The simple beta comes from a regression on the current market return; the summed beta adds the coefficients of three lagged market returns. Short Selling, with a beta of about −0.7, is omitted for readability. Figure produced using Matplotlib (Hunter, 2007).

Figure 2 shows that the lagged beta has been stable over time rather than a product of one episode. Estimated over rolling 60-month windows, it stays between about 0.03 and 0.15 for most of the period. The main exception is the window containing the 2008 financial crisis, when the estimate briefly rises above 0.2 before falling back as the crisis months leave the window.



Figure 2. Lagged beta of the EDHEC composite over rolling 60-month windows, with a 90% Newey–West confidence interval. Windows ending before October 2005 include months from the replication period. Figure produced using Matplotlib (Hunter, 2007).

4.3 Alpha after accounting for the lags

H3 is not supported: out of sample, hedge funds retain a positive alpha even after the lagged market exposure is counted. The lagged-model alpha of the EDHEC composite is 2.45% per year, with a 95% confidence interval from 0.68% to 4.21%, only modestly below the simple alpha of 2.84%. The result differs from the original study mainly because the market exposure is smaller and the market's return lower: an extra 0.06 of beta on a market that earned 7.5% per year above the T-bill explains only about 0.4 percentage points of return, whereas in the late 1990s an extra 0.47 of beta on a market earning 14.6% explained almost 7 points.

The alpha is not stable over time. It is 4.06% per year between October 2000 and December 2008 but only 0.49% after 2009, with a p-value of 0.70, so the full-period result is driven by the early years. For the Funds of Funds index, which reflects the returns actually available to investors in diversified hedge fund portfolios after a second layer of fees, the lagged-model alpha is 0.03% per year, and hedging with the summed beta reduces its Sharpe ratio from 0.47 to zero. The positive alpha of the composite should also be read with the survivorship and backfill biases of hedge fund indices in mind (Liang, 2000; Fung & Hsieh, 2000), which raise index returns above what investors earned.

4.4 Liquidity and asymmetry

H4 is supported. The lagged beta of an equal-weighted portfolio of the five illiquid styles is 0.10, against 0.02 for the four liquid styles, and the difference, 0.076, is significant ($p = 0.006$). This is the pattern predicted by the stale-pricing explanation: delays in reported returns are concentrated in strategies holding convertible bonds, distressed debt and other instruments without frequent market prices.

H6 is not supported, and the evidence points the other way. Asness et al. (2001) found a lagged beta of 0.79 after down markets and 0.17 after up markets for the aggregate index, and interpreted the asymmetry as a sign that managers delay the recognition of losses. Out of sample, the lagged beta of the composite is 0.16 after up markets and −0.03 after down markets, so the down-minus-up difference is −0.18 with a standard error of 0.05. The late adjustment of hedge fund returns in 2000–2021 followed market rises, not market falls. This does not rule out deliberate smoothing, which Bollen and Pool (2008, 2009) documented at the level of individual funds, but it means that the asymmetry test gives no support to that interpretation at the index level in this period. A plausible alternative is that sharp market declines, such as those of late 2008 and March 2020, are marked quickly even in illiquid books, whereas gradual recoveries in the prices of hard-to-value assets are recognised with a delay.

4.5 The 2022–2024 extension

H5 is supported. On the Barclay Hedge Fund Index, the lagged beta over January 2022 to December 2024 is 0.055 with a standard error of 0.031 ($p = 0.039$), almost exactly the size found in the EDHEC data for 2000–2021. With only 36 months, the estimate is imprecise, and

its 90% interval runs from 0.004 to 0.106. The more striking feature of this period is the size of the ordinary market exposure. The simple beta of the Barclay index is 0.35 and its correlation with the stock market 0.94, and its summed beta is 0.42. Its alpha is negative, at −2.90% per year, because the index lost money in 2022 along with the stock market and then gained less than the market in 2023 and 2024. This comparison uses a different index family, built as an average of all reporting funds rather than of style indices, so its level is not directly comparable with the EDHEC composite, but it shows that the lagged exposure was still present in the most recent data available.

4.6 Robustness

The primary result is robust to the pre-specified variations. Table 7 summarises the 144 specifications, and Figure 3 shows the distribution of the estimates. All 144 lagged betas are positive, ranging from 0.02 to 0.09 with a median of 0.062, and 127 are significant at the 5% level. All 17 non-significant specifications combine six lags with the EDHEC composite; adding three further monthly coefficients raises the standard error without adding to the estimate, since the fourth to sixth lags are close to zero. The result does not depend on the 2008 crisis, on the March 2020 crash or on controlling for the size, value and momentum factors. Equally clearly, none of the 144 estimates reaches the smallest effect of interest of 0.10: no reasonable specification recovers a lagged exposure of the size found before 2000.

Table 7. Robustness grid: the primary test (H2) across 144 specifications.

Choice	Option	Specifications	Median lagged beta	Range	Significant at 5%
Number of lags	1	36	0.06	0.04 to 0.06	36 of 36
	2	36	0.07	0.06 to 0.09	36 of 36
	3	36	0.06	0.04 to 0.08	36 of 36
	6	36	0.05	0.02 to 0.08	19 of 36
Standard errors	Newey–West, 3 lags	48	0.06	0.02 to 0.09	42 of 48
	Newey–West, 6 lags	48	0.06	0.02 to 0.09	42 of 48
	OLS	48	0.06	0.02 to 0.09	43 of 48
Index	EDHEC composite	72	0.05	0.02 to 0.07	55 of 72
	Funds of Funds	72	0.07	0.05 to 0.09	72 of 72
Sample	Without Sep–Dec 2008	48	0.06	0.02 to 0.09	42 of 48
	Without Feb–Apr 2020	48	0.06	0.04 to 0.09	43 of 48
	Full	48	0.06	0.03 to 0.09	42 of 48
Controls	SMB, HML, momentum	72	0.06	0.02 to 0.08	63 of 72
	None	72	0.07	0.03 to 0.09	64 of 72

Note: Each row pools all specifications sharing that option; October 2000 – May 2021. With six lags the lagged beta is β1 + … + β6. One-sided tests. Source: Author's own calculations.

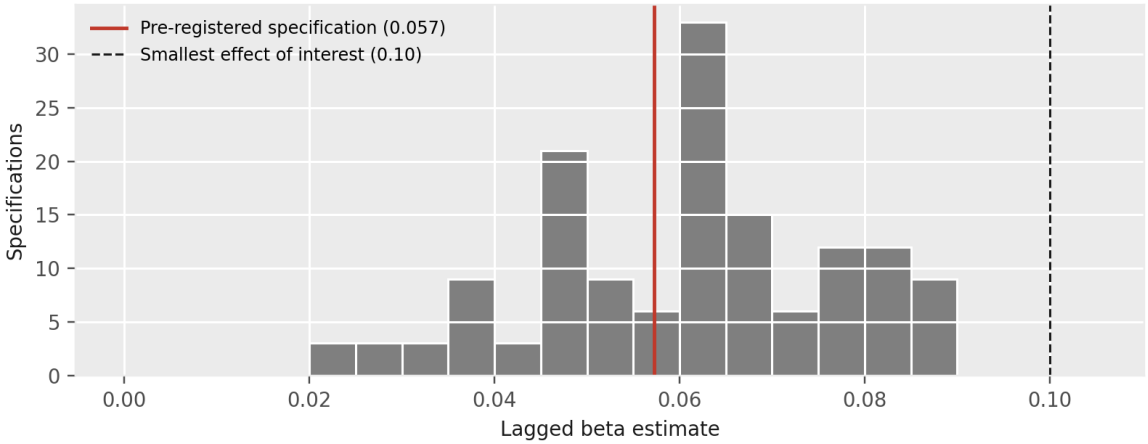


Figure 3. Distribution of the lagged beta across the 144 pre-specified robustness specifications, October 2000 – May 2021. Figure produced using Matplotlib (Hunter, 2007).

The independent re-implementation reproduces the primary estimate, standard error and p-value exactly, the look-ahead check passes, and the data checks confirm the dating of the EDHEC file and the absence of any observation after December 2024.

5. Discussion

5.1 Do hedge funds still hedge?

The short answer is that hedge funds, taken as a group, hedge more than they did in the late 1990s, but still less than their reported returns suggest. The replication confirms that the finding of Asness et al. (2001) was not an artefact of their data: with a different index provider, a different market index and a partly different period, lagged market returns again reveal a large hidden exposure before 2000. After 2000, the hidden exposure is still there, statistically robust and concentrated in the least liquid strategies, but it is much smaller, adding about 0.06 to a beta of 0.14. An investor who measures a hedge fund index's market exposure with the usual monthly regression will still underestimate it, but by about 40% rather than by more than half.

Several developments since 2000 could explain the decline. Valuation practices were tightened after the losses of 2008, with more independent administrators and pricing controls, which Cassar and Gerakos (2011) link to less smoothing. The composition of the industry also shifted towards institutional investors, who demand more frequent and more reliable valuations. The persistence of the lag in distressed securities, event-driven and fixed-income arbitrage funds shows, however, that illiquid assets still produce stale reported returns, as the model of Getmansky et al. (2004) predicts.

5.2 What this means for investors and for hedge fund failures

For investors, the practical lesson is that monthly statistics of hedge fund returns flatter them. Positive autocorrelation reduces measured volatility and raises Sharpe ratios (Lo, 2002), and lagged exposure lowers measured betas and correlations. Both make hedge funds look like better diversifiers than they are, and both matter most in a crisis, when the delayed exposure is eventually recognised. In 2022, when stocks and bonds fell together, the Barclay index lost 8.2%, and its monthly returns moved closely with those of the stock market.

The results also put the question raised in the introduction into perspective. It is tempting to say that hedge funds implode because they do not hedge. The evidence here supports a more precise version of that statement. At the level of the industry, the typical hedge fund does hedge part of its market risk, but less than it appears to, because some of the exposure only shows up in reported returns with a delay. The famous collapses were different in kind: they were caused by individual funds taking large, leveraged and concentrated positions that were either unhedged, as at Amaranth and Archegos, or hedged in ways that failed under stress, as at LTCM. Index-level regressions cannot detect such risks, which lie in the tails of individual funds rather than in the average.

5.3 Limitations

The study has several limitations. First, the data are indices, not investable funds or the returns investors actually earned, and they are subject to survivorship and backfill biases that

raise average returns (Liang, 2000; Fung & Hsieh, 2000; Dichev & Yu, 2011); the alphas reported here should be read as upper bounds. Second, the EDHEC indices are averages of other providers' indices, and averaging could itself smooth returns, although this would not create a relation with lagged market returns unless the underlying indices already had one. Third, the replication covers only 45 of the original 81 months, because free hedge fund data begin in 1997, and it uses a different index provider and market index. Fourth, the extension relies on a different index family, covers only 36 months and was transcribed from a public web page; the months from June to December 2021 are not covered. Fifth, a positive lagged beta is consistent with stale or smoothed prices, but also with a genuine delayed reaction of hedge fund strategies to the market, as in merger arbitrage (Mitchell & Pulvino, 2001); like the original study, this one cannot fully separate the two, although the concentration of the lag in illiquid strategies favours the pricing explanation. Finally, the tests measure average linear exposures and say nothing about the option-like crash risk documented by Agarwal and Naik (2004).

6. Conclusion

Twenty-four years after Asness, Krail and Liew (2001) asked whether hedge funds hedge, this study repeated their test under a pre-registered protocol on data they could not have seen. The original finding replicates: in 1997–2000, lagged market returns triple the measured market beta of hedge fund indices and remove most of their apparent alpha. Out of sample, from October 2000 to May 2021, the lagged exposure is still statistically significant, robust across 127 of 144 specifications and concentrated in illiquid strategies, and it is still present in 2022–2024. Its size, however, has fallen by about two-thirds, and it no longer removes the hedge fund alpha over the full period, although that alpha has been close to zero since 2009 and for funds of funds. The asymmetry that the original study interpreted as evidence of deliberate smoothing is reversed.

Hedge funds therefore hedge more than they did, but still less than their reported returns suggest. Investors should adjust hedge fund betas, correlations and Sharpe ratios for stale pricing before allocating to them, particularly for credit, distressed and arbitrage strategies. The same method could be applied to other assets with infrequent valuations, such as private equity, private credit and real estate funds, whose reported returns are likely to hide much larger lagged exposures than those found here.

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Appendixes

Appendix A. Lagged-beta regressions for every EDHEC index, October 2000 – May 2021.

Index	Months	Simple beta	Lagged beta	(t)	p	p (Holm)	Summed beta	Simple alpha (%/yr)	Lagged-model alpha (%/yr)
Convertible Arbitrage	248	0.21	0.07	(1.11)	0.133	0.665	0.27	2.89	2.43
CTA Global	248	−0.01	−0.02	(−0.42)	0.664	0.770	−0.03	3.67	3.82
Distressed Securities	248	0.26	0.17	(5.73)	<0.001	<0.001	0.43	4.40	3.19
Emerging Markets	248	0.46	0.09	(1.97)	0.024	0.169	0.55	3.06	2.45
Equity Market Neutral	248	0.08	0.05	(2.11)	0.018	0.141	0.13	1.80	1.44
Event Driven	248	0.33	0.10	(3.20)	<0.001	0.008	0.42	3.15	2.49
Fixed Income Arbitrage	248	0.12	0.10	(3.10)	<0.001	0.010	0.21	3.04	2.40
Global Macro	248	0.13	0.01	(0.41)	0.341	0.770	0.14	3.27	3.18
Long/Short Equity	248	0.39	0.05	(1.68)	0.047	0.281	0.44	1.69	1.33
Merger Arbitrage	248	0.15	0.02	(0.87)	0.193	0.770	0.17	2.74	2.57
Relative Value	248	0.21	0.06	(2.50)	0.006	0.056	0.26	2.84	2.45
Short Selling	241	−0.68	0.04	(0.69)	0.246	0.770	−0.64	0.02	−0.38
Funds of Funds	248	0.23	0.08	(3.26)	<0.001	–	0.31	0.57	0.03
EDHEC composite	248	0.14	0.06	(2.36)	0.009	–	0.20	2.84	2.45

Note: As in Table 6. t-statistics and one-sided p-values from Newey–West standard errors (3 lags); Holm adjustment across the 12 style indices. Short Selling excludes the seven months the distributor zero-filled. Source: Author's own calculations.

Appendix B. Lagged-beta regressions for every EDHEC index, January 1997 – September 2000.

Index	Months	Simple beta	Lagged beta	(t)	p	p (Holm)	Summed beta	Simple alpha (%/yr)	Lagged-model alpha (%/yr)
Convertible Arbitrage	45	0.06	0.24	(2.61)	0.005	0.035	0.32	7.21	2.95
CTA Global	45	0.01	−0.02	(−0.15)	0.558	0.743	−0.02	1.13	1.65
Distressed Securities	45	0.26	0.31	(2.73)	0.003	0.028	0.60	1.60	−4.04
Emerging Markets	45	0.74	0.41	(1.59)	0.055	0.172	1.19	−7.35	−14.76
Equity Market Neutral	45	0.09	0.08	(2.62)	0.004	0.035	0.18	6.63	5.23
Event Driven	45	0.29	0.25	(2.49)	0.006	0.038	0.57	4.11	−0.46
Fixed Income Arbitrage	45	−0.01	0.34	(2.31)	0.011	0.053	0.36	0.52	−5.44
Global Macro	45	0.29	0.27	(2.85)	0.002	0.022	0.58	3.28	−1.55
Long/Short Equity	45	0.37	0.19	(3.17)	<0.001	0.009	0.58	8.85	5.42
Merger Arbitrage	45	0.15	0.10	(1.72)	0.043	0.172	0.26	7.34	5.53
Relative Value	45	0.11	0.18	(2.93)	0.002	0.019	0.30	6.32	3.09
Short Selling	45	−1.21	0.08	(0.33)	0.372	0.743	−1.13	16.82	15.47
Funds of Funds	45	0.33	0.29	(3.23)	<0.001	–	0.64	4.88	−0.29
EDHEC composite	45	0.10	0.20	(3.07)	0.001	–	0.32	4.71	1.09

Note: As in Table 6. t-statistics and one-sided p-values from Newey–West standard errors (3 lags); Holm adjustment across the 12 style indices. Short Selling excludes the seven months the distributor zero-filled. Source: Author's own calculations.

Appendix C. Annualised Sharpe ratios before and after hedging the market exposure.

Period	Index	Unhedged	Hedged with simple beta	Hedged with summed beta
1997–2000	EDHEC composite	1.76	1.52	0.26
	Funds of Funds	1.26	0.90	0.00
	Convertible Arbitrage	2.02	1.86	0.53
	CTA Global	0.16	0.14	0.20
	Distressed Securities	0.82	0.31	−0.47
	Emerging Markets	0.23	−0.59	−0.97
	Equity Market Neutral	3.25	3.57	2.22
	Event Driven	1.22	0.85	−0.03
	Fixed Income Arbitrage	0.08	0.09	−0.60
	Global Macro	0.96	0.52	−0.16
	Long/Short Equity	1.79	1.79	0.91
	Merger Arbitrage	2.26	2.19	1.45
	Relative Value	2.38	2.27	0.74
	Short Selling	−0.08	1.08	0.99
2000–2021	EDHEC composite	1.06	0.97	0.80
	Funds of Funds	0.47	0.17	0.00
	Convertible Arbitrage	0.73	0.57	0.46
	CTA Global	0.46	0.47	0.49
	Distressed Securities	1.02	0.95	0.59
	Emerging Markets	0.67	0.47	0.37
	Equity Market Neutral	0.89	0.75	0.57
	Event Driven	0.86	0.78	0.58
	Fixed Income Arbitrage	1.06	0.95	0.68
	Global Macro	1.00	0.88	0.86
	Long/Short Equity	0.66	0.50	0.38
	Merger Arbitrage	1.02	0.92	0.86
	Relative Value	1.05	1.07	0.87

Note: Following Asness et al. (2001, Exhibit 6), hedged returns subtract the in-sample beta times the market excess return each month. Short Selling is omitted in 2000–2021 because of missing months. Source: Author's own calculations.